

untangle::executor

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Chapter 1

executor

c++ thread pool executor based on async execution

A header-only library: [executor.hpp](#) exposes a pool of workers, each one an [async](#) execution in continuous mode, fed by a shared queue of tasks.

requirements

C++23, and the async submodule:

```
git submodule update --init --recursive
```

async carries its own actuator submodule, which is why the clone has to be recursive. Both are header-only; nothing is built or installed.

The headers report warnings with `std::println`, so they need a standard library that provides `<print>` — GCC 14 or newer.

example

```
#include <executor.hpp>
#include <functional>

int main() {
    untangle::executor<std::function<void(void)>> pool(4);
    pool.start();

    for (int i = 0; i < 100; ++i) {
        pool.add_task([i] { work(i); }, [] { /* finished */ });
    }

    return 0; // ~executor finishes what was added, then stops the workers
}
```

the action type

The pool is a template on the callable it runs, the way an execution is — and on the same parameter name, because every task queued is that one callable. One pool runs one signature, and the caller names it:

```
untangle::executor<std::function<int(void)>> pool(4); // each result goes to the task's callback
```

The action type does not have to be a `std::function`. What an execution requires of it is a nested `result_type`, which it reads rather than deduces, so any callable declaring one will do — and a pool built on such a callable never touches `std::function::result_type`, which the standard removed in C++20 and both `libc++` and `libstdc++` still supply by grace rather than contract.

```
struct my_task {
    using result_type = void;
    void operator()() const;
};

untangle::executor<my_task> pool(4);
```

tasks: work that reports back

Work enters the pool through one door, `add_task()`, and every task carries the callback it must notify. An action type may take arguments, and `add_task()` binds them the way its namesake `execution::add_task()` does — at the door, where the caller still holds them — with the callback after them:

```
untangle::executor<std::function<int(int)>> pool(4);
pool.start();

// the callback comes last, after the arguments the task is bound to
pool.add_task([](int n) { return n * 2; }, 21, [](int result) { /* result == 42 */ });
```

What waits for a worker is one callable carrying its own arguments and its callback, so nothing about a task changes between the queue and the worker that runs it. The arguments are copied at `add_task()` and handed to the task as the pool's own lvalues when it runs: a task is called later, possibly much later, and what the caller passed may be gone by then. A task taking `int&` therefore mutates the pool's copy, which the caller never sees.

A task that cannot be called with the arguments given is refused by a `static_assert` in `add_task()`, which names the mismatch before the errors from inside the binding follow it.

Work that returns `void` still reports that it finished, with a callback taking nothing — finished is the message and the result is optional:

```
untangle::executor<std::function<void(void)>> pool(4);
pool.add_task([] { /* ... */ }, [] { /* finished */ });
```

A task with nothing to bind passes the callback alone, whatever it returns:

```
untangle::executor<std::function<int(void)>> pool(4);
pool.add_task([] { return 42; }, [](int result) { /* result == 42 */ });
```

The callback is the last argument, and the signature cannot say so. A parameter pack cannot be followed by a deducible parameter, so it arrives inside the arguments and `untangle::bind_task()` splits it off. It must be callable with the task's result and return nothing — or callable with nothing at all, when the task returns `void`. A missing or unusable one is a compile error rather than silence.

Work nobody wants told about still passes a callback, one that ignores what it is given. There is no fire-and-forget door: a `void` task's callback costs one empty call, and a single door means a single queue, in which the order out is the order in.

```
constexpr auto ignore_result = [](auto&&...) {}; // takes a result, or nothing for a void task
pool.add_task([](int n) { return n * 2; }, 21, ignore_result);
```

A refused task does not notify. `add_task()` answers false before `start()`, after `stop()`, while a `wait()` is draining, once the destructor has begun, or if the worker it was offered to has stopped — and a refused

task is destroyed rather than queued. The answer is the whole report, so a caller waiting on the callback rather than on the answer would wait for ever. The same is true of work still queued when a stopped pool is destroyed: it never runs, so it never notifies, and the count on stderr is all there is.

Finished does not mean failed. A task that throws is not notified: there is no result to hand over, and for a void task no completion to report either. What it threw goes to `on_task_error`. And because a callback runs inside the same try as the task that owns it, `on_task_error` can fire for a task that in fact succeeded — the work was done and only the telling failed.

A callback runs on a worker's thread, inside that worker's drain, and several may be running at once on different workers. It is the same rule `on_task_error` carries, for the same reason: nothing may escape it.

how work is placed

N executions run in continuous mode behind a shared `std::deque` of tasks.

Nothing overtakes. A task goes straight to a free worker only when the shared queue is empty; otherwise it joins the back of the queue. Finding a free worker does not let a task jump a queue that already has work in it.

Workers pull rather than being pushed to. When one reports its queue finished — `on_finished`, raised by execution the moment its list empties — it takes the front of the shared queue. That is what spreads the work, with no scheduling logic of its own: 400 tasks over 4 workers come out 100/100/100/100.

A worker is asked, not tracked. `execution::is_busy()` is what `add_task()` reads to find a free worker and what the destructor reads to decide the pool is idle. A tally kept here would be the pool's belief about its workers; this is the workers' own answer, and it cannot drift.

running, stopping, restarting

A pool is built stopped. The constructor makes the workers and wires them up; `start()` runs them. Until it is called the pool refuses work, because a task taken then would sit unrun and the destructor would wait for it. It is also the moment to assign `on_task_error`, before anything can throw.

`stop()` stops the workers, and stops the pool accepting. What a worker is already running it finishes; what is still queued stays queued. `stop()` does not wait for the workers to leave their threads — only the destructor does that, and that wait is what makes destroying the pool safe.

`start()` after `stop()` restarts it, workers and all, and the queue it was holding drains as they pick it up again. Either call is harmless twice over.

`wait()` is a barrier, not a shutdown. It returns once everything added has run and no worker is busy, and it leaves the pool exactly as it found it — workers still in their threads, nothing destroyed, accepting again. Work added after it runs as it did before, and calling it repeatedly is how you step through work in phases:

```
pool.add_task([ { first(); }, [] {}]);
pool.wait();
pool.add_task([ { second(); }, [] {}]);
pool.wait();
```

// first() has run, the pool is still alive
// and so has second()

It refuses new work while it waits, then restores accepting to whatever it was. A task that re-adds itself would otherwise keep the drain from ever ending, so `add_task()` answers false for the duration — and because the previous state is restored rather than assumed, `wait()` on a stopped pool does not leave it accepting.

So only work accepted before the wait began is what the wait covers. Work offered during it was never queued, and work added after it returns is simply the next batch, for the next `wait()`. This is a stop-the-world drain, not a transparent one: a producer on another thread sees `add_task()` answer false while a `wait()` is in progress, and has to offer the work again afterwards.

A pool that is not running is not waited for. Only a worker takes from the queue, so a stopped pool can never empty it, and a wait insisting otherwise would never return. `wait()` returns at once instead, leaving what is queued for the next `start()`.

lifetime and failure

The destructor finishes what was added. Its drain is `wait()`, so a task added before the pool goes out of scope has run by the time it does. What the destructor adds is the shutdown `wait()` deliberately leaves out, in the order that makes it safe: stop the workers, wait for every one of them to leave its thread, and only then destroy them.

The destructor waits for the pool, not for what the pool's work touched. That distinction is the whole reason `wait()` is public. The drain guarantees no worker is still running when the pool is destroyed, which is a statement about the pool's own members — a destructor body runs before them, so they are all still alive for it. It says nothing about the lifetime of what a task captured.

For a pool that is a local variable outliving what its work touches, the destructor is enough and nothing else is needed. For a pool held as a member of another class, it is already too late: `~executor()` runs during its owner's member destruction, so by then the owner's derived subobject is gone, and so is every member declared after the pool. A callback reaching into either of those runs against destroyed storage however faithfully the pool drains — and on the stack that is not even a fault a sanitizer can name, because the memory is still addressable.

So an owner whose callbacks reach back into itself calls `wait()` where the whole object graph is still standing — in its own destructor body, ahead of its members, or better still at a point the owner of its peers controls — and the destructor's drain goes back to being a backstop rather than the thing relied upon.

A pool destroyed while stopped is the exception: its queue can never empty, because only a running worker takes from it, so the destructor does not wait for it. Whatever is left is reported on `stderr` and dropped.

Both waits are unbounded, and both say so while they last. A task that never returns would otherwise leave the destructor silent for ever, so it reports instead — the first wait naming who is still inside a task, the second naming whose thread has not left. The first line below is the drain's, so a plain `wait()` prints it exactly as a destructor does; the second is the destructor's alone:

```
executor: still waiting to drain after 1s - 1 queued, 2 in a task: pool_worker_0, pool_worker_1
executor: still waiting to drain after 3s - 0 queued, 1 in a task: pool_worker_1
executor: still waiting to stop after 1s - 1 not left its thread: pool_worker_1
```

The interval starts at a second and doubles to a ceiling of thirty, so a stuck teardown says something almost at once without a long one becoming a flood. The two waits are kept apart because they answer different questions: a worker can be idle and still be in its thread.

Bounding either wait was never available. Returning early while a worker is still in a task is a use-after-free, and a bounded wait only moves the hang into `~execution()`, which spins on its own running state with no timeout. Workers are detached and `async::execution` has no cancellation, so a pool cannot outlive a task it cannot interrupt. What can be fixed is the silence.

`add_task()` says whether the task was taken. It returns false before `start()`, after `stop()`, while a `wait()` is draining, and once the destructor has begun. It returns false again if the worker it was offered to has stopped. A refused task is destroyed rather than queued, so a false answer means it will not run.

A task that is still running is refused too, and that is chosen rather than incidental. A pool draining is waiting for exactly that task, so it is tempting to let the work it spawns through — but then a task that re-adds itself could keep the drain from ever ending, and that holds for a `wait()` just as for a destructor. One rule instead: a pool that is not running takes no work, whoever is asking. `add_task()` answers false inside the task, so it can tell.

A task that throws does not take the worker with it, and what it threw is not lost. The worker catches it and carries on with the next task; `on_task_error` is where the throw goes:

```

untangle::executor<std::function<void(void)» pool(4);

pool.on_task_error = [](std::exception_ptr thrown) {
    try {
        std::rethrow_exception(thrown);
    } catch (const std::exception& e) {
        log(e.what());
    }
};

```

It carries a `std::exception_ptr` because a task may throw something that is not a `std::exception`, and that is the case most worth hearing about. Assign it before the first `add_task()` — a worker reads it — and expect it on a worker's thread, with more than one worker possibly inside it at once. Leave it unset and the pool warns on `stderr` instead, naming the worker.

That name is not unique between pools. Workers are named by index — `pool_worker_0`, `pool_worker_1` — so two pools in one process each have a `pool_worker_0`, and every warning that names one is ambiguous: the failure reports above, and the destructor's `wait` alike. With a single pool the name says which worker; with two it does not. Assigning `on_task_error` sidesteps it for failures, since the handler is the pool's own and the warning is replaced.

A return value is not collected by the pool: a continuous worker keeps none. It goes to the task's own callback — see `tasks: work` that reports back above.

building the tests

```

cmake -S test -B test/build
cmake --build test/build
ctest --test-dir test/build

```

`test/CMakeLists.txt` fetches googletest at configure time, so the first configure needs network access.

sanitizers

`EXECUTOR_SANITIZE` builds the tests under a sanitizer. It is off by default, because a sanitized build is several times slower and ThreadSanitizer does not ship for every toolchain.

```

cmake -S test -B test/build-asan -DEXECUTOR_SANITIZE=address
cmake --build test/build-asan
ctest --test-dir test/build-asan

```

Accepted values are `address`, `thread`, `undefined`, or `empty`. Anything else is refused at configure time rather than passed through to the compiler.

Use a separate build directory per sanitizer. `address` and `thread` instrument the same accesses in incompatible ways and cannot be combined, so one build directory is one sanitizer.

the reference

`tools/make_doc.sh` builds `doc/refman.pdf` from the header and this README. It needs `doxygen`, `tectonic` and `python3`, and it fails on any `doxygen` warning that is not an `obsolete-tag` notice, so an undocumented member or a broken `\ref` does not reach the reference.

Chapter 2

Namespace Index

2.1 Namespace List

Here is a list of all documented namespaces with brief descriptions:

untangle	A thread pool executor built out of <code>untangle::async::execution</code> objects	13
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Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

untangle::executor< actionT >	
Runs tasks on a fixed pool of untangle::async::execution workers	15

Chapter 4

File Index

4.1 File List

Here is a list of all documented files with brief descriptions:

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--	----

Chapter 5

Namespace Documentation

5.1 untangle Namespace Reference

A thread pool executor built out of `untangle::async::execution` objects.

Classes

- class `executor`
Runs tasks on a fixed pool of `untangle::async::execution` workers.

5.1.1 Detailed Description

A thread pool executor built out of `untangle::async::execution` objects.

N workers in continuous mode behind a shared queue. Nothing overtakes what is already queued.

Chapter 6

Class Documentation

6.1 `untangle::executor< actionT >` Class Template Reference

Runs tasks on a fixed pool of `untangle::async::execution` workers.

```
#include <executor.hpp>
```

Public Member Functions

- `executor` (`std::size_t worker_count`)
Builds `worker_count` workers. They do not run until `start()`.
- `~executor` ()
Drains what was added, then stops the workers and destroys them.
- `void wait` ()
Drains everything added so far, then leaves the pool as it was found.
- `template<typename... Args>`
`bool add_task` (`actionT task`, `Args &&... args`)
Queues work that must report, or hands it to a worker that has nothing to do.
- `std::size_t pending` () `const`
How many tasks are waiting for a worker.
- `void start` ()
Starts every worker, and starts the pool accepting work.
- `void stop` ()
Stops every worker. What they are already running, they finish.

Public Attributes

- `std::function< void(std::exception_ptr)> on_task_error`
Called with whatever the work threw. Assigned by the caller.

6.1.1 Detailed Description

```
template<typename actionT>  
class untangle::executor< actionT >
```

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Runs tasks on a fixed pool of `untangle::async::execution` workers.

`add_task()` is the one door: it carries a callback and reports the result to it.

Template Parameters

actionT	- The callable the pool runs. One pool runs one signature, arguments and result.
---------	--

6.1.2 Constructor & Destructor Documentation

6.1.2.1 executor()

```
template<typename actionT>
untangle::executor< actionT >::executor (
    std::size_t worker_count) [inline], [explicit]
```

Builds worker_count workers. They do not run until [start\(\)](#).

Parameters

worker_count	- How many workers to run. Must not be zero.
--------------	--

Exceptions

std::invalid_argument	- worker_count is zero: such a pool could never run its work.
-----------------------	---

6.1.2.2 ~executor()

```
template<typename actionT>
untangle::executor< actionT >::~~executor () [inline]
```

Drains what was added, then stops the workers and destroys them.

The drain is [wait\(\)](#); the shutdown after it stops the workers, waits for each to leave its thread, and destroys them only then. Both waits are unbounded and report on stderr meanwhile.

Attention

Only a pool stopped before its queue emptied leaves work behind, and work dropped here never runs, so it never notifies. The count on stderr says how much, never which.

This waits for the pool, not for what its work touched: see [wait\(\)](#).

6.1.3 Member Function Documentation

6.1.3.1 wait()

```
template<typename actionT>
void untangle::executor< actionT >::wait () [inline]
```

Drains everything added so far, then leaves the pool as it was found.

Drained means the queue is empty and no worker is busy, so every task accepted before this has run, and so has the callback of one - a callback runs inside the worker's drain, not after it.

A barrier, not a shutdown: the workers keep their threads and work added afterwards runs as before. Reusable, and one at a time - a second caller drains once the first has returned.

The wait is unbounded and reports on stderr while it lasts.

Attention

New work is refused meanwhile, then accepting is restored, so a task that re-adds itself cannot keep the drain from ending. Only work accepted before it began counts.

A pool that is not running is not waited for: only a worker takes from the queue, so this returns at once and what is queued keeps for the next [start\(\)](#).

It waits for the work, not for what the work touched. Held as a member, [~executor\(\)](#) runs once the owner's derived parts and later members are gone: call this before that.

6.1.3.2 add_task()

```
template<typename actionT>
template<typename... Args>
bool untangle::executor< actionT >::add_task (
    actionT task,
    Args &&... args) [inline]
```

Queues work that must report, or hands it to a worker that has nothing to do.

The first free worker takes it, the warmest one; otherwise it waits in the queue, which keeps the order work arrived in. Once the task has run, its callback is invoked with the result, on the worker's thread.

Attention

The last argument is the callback, which the signature cannot say: it arrives in args and [untangle::bind_task\(\)](#) splits it off. A missing or unusable one is a [static_assert](#), and so is a task that cannot be called with the arguments before it.

A task that throws is not notified - what it threw goes to [on_task_error](#) - and a refused one does not notify either, so a caller waiting on the callback alone waits for ever.

An empty callback or action is refused here too. Only a run time check can see it, and a task that can neither run nor notify is turned away rather than queued to throw.

Template Parameters

Args	- The argument types to bind, of which the last is the callback.
------	--

Parameters

task	- The task to run.
args	- What to bind, followed by the callback. Copied here, so a task taking a reference gets the pool's copy.

Returns

true - accepted, and the callback will be notified once it has run.

false - refused by a pool not started, stopped, draining or being destroyed, by a stopped worker, or because the task could never do its job. It will not run or notify.

6.1.3.3 start()

```
template<typename actionT>
void untangle::executor< actionT >::start () [inline]
```

Starts every worker, and starts the pool accepting work.

A pool refuses work until this is called, and again after [stop\(\)](#) until it is called anew, which revives the workers too. Harmless twice over. [wait\(\)](#) needs nothing from here.

6.1.3.4 stop()

```
template<typename actionT>
void untangle::executor< actionT >::stop () [inline]
```

Stops every worker. What they are already running, they finish.

Queued tasks stay unrun and anything added afterwards is refused, a running task included. [start\(\)](#) revives the workers; harmless twice over.

Attention

It does not wait for the workers to leave their threads - only the destructor does, and that wait is what makes destroying the pool safe.

6.1.4 Member Data Documentation

6.1.4.1 on_task_error

```
template<typename actionT>  
std::function<void(std::exception_ptr)> untangle::executor< actionT >::on_task_error
```

Called with whatever the work threw. Assigned by the caller.

The only report of failed work, since [add_task\(\)](#) answered long before it ran. Left unset, the pool warns on stderr. Rethrow the exception_ptr to read it.

Remarks

A task's failure arrives here, and so does its callback's - so this can fire for a task that succeeded, when only the telling failed.

Attention

It runs on a worker's thread, several at once. Assign it before the first [add_task\(\)](#), and let nothing escape it or a callback.

The documentation for this class was generated from the following file:

- executor.hpp

Chapter 7

File Documentation

7.1 executor.hpp

```
00001 // Copyright (c) 2026 Nicolae Popescu. MIT License.
00002
00008 #ifndef UNTANGLE_EXECUTOR_EXECUTOR_HPP
00009 #define UNTANGLE_EXECUTOR_EXECUTOR_HPP
00010
00011 #include <algorithm>
00012 #include <async.hpp>
00013 #include <atomic>
00014 #include <chrono>
00015 #include <condition_variable>
00016 #include <cstdint>
00017 #include <deque>
00018 #include <exception>
00019 #include <functional>
00020 #include <memory>
00021 #include <mutex>
00022 #include <print>
00023 #include <stdexcept>
00024 #include <string>
00025 #include <thread>
00026 #include <tuple>
00027 #include <type_traits>
00028 #include <utility>
00029 #include <vector>
00030
00031 namespace untangle {
00032
00040 template <typename actionT>
00041 class executor {
00042 public:
00049 explicit executor(std::size_t worker_count) {
00050     if (worker_count == 0) {
00051         throw std::invalid_argument("executor: worker_count must not be zero");
00052     }
00053
00054     workers_.reserve(worker_count);
00055
00056     for (std::size_t index = 0; index < worker_count; ++index) {
00057         auto next = worker_execution::create_instance("pool_worker_" + std::to_string(index));
00058
00059         // Tells the pool the moment this worker has room for more.
00060         next->on_finished = [this, index] { take_next_task(index); };
00061
00062         // Carries what a task throws to the caller. Assigned here because a worker thread reads it.
00063         next->on_error = [this, name = next->name](std::exception_ptr thrown) {
00064             report_task_error(name, thrown);
00065         };
00066
00067         // Adds the worker to the poll the destructor waits on.
00068         poll_.add(*next);
00069
00070         workers_.push_back(std::move(next));
00071     }
00072 }
00073
00085 ~executor() {
00086     // Nothing is stopped before this, or the queue could never empty. wait() leaves the pool
```

```

00087 // accepting, so stop() follows at once: that window is the only one a shutdown would take.
00088 wait();
00089 stop();
00090
00091 {
00092     std::lock_guard<std::mutex> lock(mutex_);
00093
00094     // Only reachable for a pool that was already stopped. wait() drained a running one, and
00095     // stop() above cannot add to the queue.
00096     if (!pending_.empty()) {
00097         std::println(stderr,
00098             "executor: {} queued task(s) dropped, the pool was stopped before they ran",
00099             pending_.size());
00100     }
00101 }
00102
00103 // Waits for every worker to leave its thread, which is what makes destroying them safe. A
00104 // different question from the wait above: a worker can be idle and still be in its thread.
00105 auto interval = std::chrono::milliseconds(report_first_ms);
00106 auto waited = std::chrono::milliseconds(0);
00107
00108 while (poll_.is_running()) { // time of check
00109     const auto tick = std::chrono::milliseconds(poll_interval_ms);
00110     std::this_thread::sleep_for(tick);
00111     waited += tick;
00112
00113     if (waited >= interval) {
00114         std::println(stderr, "executor: still waiting to stop after {}s - {} not left its thread{}",
00115             waited.count() / 1000, count_workers(&worker_execution::is_running),
00116             name_workers(&worker_execution::is_running));
00117
00118         interval = std::min(interval * 2, std::chrono::milliseconds(report_max_ms));
00119     }
00120 }
00121
00122 // time of use: no worker thread is left to reach workers_.
00123 workers_.clear();
00124 }
00125
00126 void wait() {
00127     // One drain at a time: two threads between the exchange below and the restore at the end would
00128     // read each other's cleared flag, and the last one out could leave the pool refusing work.
00129     std::lock_guard<std::mutex> drain(wait_mutex_);
00130
00131     // Put back exactly as found below: a wait must not leave a stopped pool accepting, and such a
00132     // pool cannot empty its queue anyway - only a running worker takes from it.
00133     const bool was_running = running_.exchange(false);
00134
00135     {
00136         std::unique_lock<std::mutex> lock(mutex_);
00137
00138         // Waits for the queue to empty and every worker to go idle.
00139         auto interval = std::chrono::milliseconds(report_first_ms);
00140         auto waited = std::chrono::milliseconds(0);
00141
00142         while (!finished_cv_.wait_for(lock, interval, [this, was_running] {
00143             return nothing_running() && (pending_.empty() || !was_running);
00144         })) {
00145             waited += interval;
00146
00147             // Under the lock, so the counts agree with the predicate that just failed.
00148             std::println(
00149                 stderr, "executor: still waiting to drain after {}s - {} queued, {} in a task{}",
00150                 waited.count() / 1000, pending_.size(), count_workers(&worker_execution::is_busy),
00151                 name_workers(&worker_execution::is_busy));
00152
00153             interval = std::min(interval * 2, std::chrono::milliseconds(report_max_ms));
00154         }
00155     }
00156
00157     running_ = was_running;
00158 }
00159
00160 template <typename... Args>
00161 bool add_task(actionT task, Args&&... args) {
00162     // The call the pool will make, which is not quite the one written at the call site: what
00163     // reaches the task is the copies bound below, so that is the call that has to compile. Without
00164     // this the mismatch surfaces deep inside bind_task()'s std::apply.
00165     static_assert(invocable_with_bound<Args...>(),
00166         "executor: the task cannot be called with the arguments given to add_task() "
00167         "before its callback");
00168
00169     if (!running_) {
00170         return false;
00171     }
00172
00173     // untangle::bind_task() takes the callback off the end of the pack and checks it; what comes

```

```

00220 // back is one callable that runs the work and then notifies, which is what the queue holds.
00221 auto tsk = untangle::bind_task(std::move(task), std::forward<Args>(args)...);
00222
00223 // bind_task() builds an empty one from an empty action or callback. Queued, it would throw on a
00224 // worker and report to on_task_error rather than to the caller, who could still act on it.
00225 if (!tsk) {
00226     return false;
00227 }
00228
00229 return queue_task(std::move(tsk));
00230 }
00231
00232 std::size_t pending() const {
00233     std::lock_guard<std::mutex> lock(mutex_);
00234     return pending_.size();
00235 }
00236
00237 void start() {
00238     // The workers first, so anything accepted after this is handed to one that is running.
00239     for (auto& one : workers_) {
00240         one->start();
00241     }
00242
00243     running_ = true;
00244 }
00245
00246 void stop() {
00247     // First, so that nothing is accepted for workers that are about to stop. A task already on a
00248     // worker still runs; one arriving from here is refused.
00249     running_ = false;
00250
00251     // No lock: a worker waking from stop() runs on_finished, which takes mutex_.
00252     for (auto& one : workers_) {
00253         one->stop();
00254     }
00255 }
00256
00257 std::function<void(std::exception_ptr)> on_task_error;
00258
00259 private:
00260 void report_task_error(const std::string& worker, std::exception_ptr thrown) {
00261     if (on_task_error) {
00262         on_task_error(thrown);
00263         return;
00264     }
00265
00266     try {
00267         std::rethrow_exception(thrown);
00268     } catch (const std::exception& e) {
00269         std::println(stderr, "warning: executor task on '{}' threw: {}", worker, e.what());
00270     } catch (...) {
00271         std::println(stderr, "warning: executor task on '{}' threw an unknown type", worker);
00272     }
00273 }
00274
00275 template <typename... Args>
00276 static constexpr bool invocable_with_bound() {
00277     if constexpr (sizeof...(Args) == 0) {
00278         return true;
00279     } else {
00280         return invocable_with<std::tuple<std::decay_t<Args>...>>(
00281             std::make_index_sequence<sizeof...(Args) - 1>{});
00282     }
00283 }
00284
00285 template <typename tupleT, std::size_t... index>
00286 static constexpr bool invocable_with(std::index_sequence<index...>) {
00287     return std::is_invocable_v<actionT, std::tuple_element_t<index, tupleT>&...>;
00288 }
00289
00290 using worker_execution = untangle::async::execution<untangle::task_t>;
00291
00292 bool queue_task(untangle::task_t task) {
00293     std::lock_guard<std::mutex> lock(mutex_);
00294
00295     // The queue comes first: an entry does not overtake one already waiting in it.
00296     if (pending_.empty()) {
00297         // Always from the start, so a free pool reuses its warmest worker rather than waking a cold
00298         // one. Under load the workers are busy and the queue below is what spreads the work.
00299         for (std::size_t index = 0; index < workers_.size(); ++index) {
00300             if (!workers_[index]->is_busy()) {
00301                 // A refused entry is already destroyed, and stop() stops every worker together, so there
00302                 // is nothing to queue and no other worker to try.
00303                 return give_to_worker(index, std::move(task));
00304             }
00305         }
00306     }
00307 }

```

```

00360
00361     pending_.push_back(std::move(task));
00362     return true;
00363 }
00364
00371 [[nodiscard]] bool give_to_worker(std::size_t index, untangle::task_t task) {
00372     // The door takes the worker's action_mutex under mutex_. The two never cycle, because a worker
00373     // calls on_finished with its action_mutex released.
00374     return workers_[index]->add_action(std::move(task));
00375 }
00376
00380 void take_next_task(std::size_t index) {
00381     std::lock_guard<std::mutex> lock(mutex_);
00382
00383     if (!pending_.empty()) {
00384         untangle::task_t next = std::move(pending_.front());
00385         pending_.pop_front();
00386
00387         if (!give_to_worker(index, std::move(next))) {
00388             // The task is already destroyed, so it is reported rather than put back.
00389             std::println(stderr, "executor: {} refused a queued task, which is lost",
00390                 workers_[index]->name);
00391         }
00392
00393         return;
00394     }
00395
00396     if (nothing_running()) {
00397         finished_cv_.notify_all();
00398     }
00399 }
00400
00406 bool nothing_running() const {
00407     for (const auto& one : workers_) {
00408         if (one->is_busy()) {
00409             return false;
00410         }
00411     }
00412
00413     return true;
00414 }
00415
00421 std::size_t count_workers(bool (worker_execution::*state)() const) const {
00422     std::size_t count = 0;
00423
00424     for (const auto& one : workers_) {
00425         if ((one.get()->*state)()) {
00426             ++count;
00427         }
00428     }
00429
00430     return count;
00431 }
00432
00439 std::string name_workers(bool (worker_execution::*state)() const) const {
00440     std::string names;
00441
00442     for (const auto& one : workers_) {
00443         if ((one.get()->*state)()) {
00444             names += names.empty() ? " : " : ", ";
00445             names += one->name;
00446         }
00447     }
00448
00449     return names;
00450 }
00451
00453 static constexpr int poll_interval_ms = 50;
00454
00456 static constexpr int report_first_ms = 1000;
00457 static constexpr int report_max_ms = 30000;
00458
00459 mutable std::mutex mutex_;
00460 std::condition_variable finished_cv_;
00461
00463 std::mutex wait_mutex_;
00464
00467 async::execution_poll poll_;
00468
00470 std::deque<untangle::task_t> pending_;
00471 std::vector<std::shared_ptr<worker_execution>> workers_;
00474 std::atomic_bool running_ = {false};
00475 };
00476
00477 } // namespace untangle
00478
00479 #endif // UNTANGLE_EXECUTOR_EXECUTOR_HPP

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